电子科技大学
格拉斯哥学院
Glasgow College, UESTC

Physical Experiment II

Lab Report

Lab Title: The Millikan Oil Drop Experiment

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Date Performed: 2023.11.2

Score:

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Abstract (About 100 words, 10 points)

The objective of this experiment is to analyze the forces on the oil drop in the stationary and motion states as well as measure the elementary charge. The Millikan oil drop apparatus and a CCD camera and a monitor are the instruments used to observe the motion of the electron under the influence of electric field force and the viscous resistance. Through this experiment and subsequent data analysis, students are able to acquire the value of elementary electron. The results obtained in this experiment is 1.602×10^{-19} C.

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Calculations and Results (Calculations, data tables and figures; 15 points)

1. Data Tables

1. DATA TABLE 4-1 (*purpose*: to measure the electric charges carried by an oil droplet)

Oil droplets	Balance voltage *V*(v)		Fall time *t*(s)	
	Measurement	Averaged	Measurement	Averaged
1	144	146	38.0	38.3
	151		38.6	
	150		38.3	
2	282	281	17.8	18.0
	281		18.0	
	281		18.2	
3	197	197	28.1	27.9
	196		27.8	
	198		28.1	
4	215	216	17..7	17.8
	216		18.0	
	217		17.7	
5	243	244	23.6	23.6
	244		23.6	
	245		23.6	

6	150	150	38.2	38.1
	150		38.1	
	150		38.1	
7	254	252	21.4	21.3
	251		21.5	
	251		21.0	
8	256	257	21.0	21.2
	257		21.2	
	258		21.4	
9	179	180	27.8	27.6
	176		27.4	
	185		27.6	
10	222	225	29.2	29.0
	224		28.6	
	227		29.3	

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2. Calculations

6. Calculations

(1) Compute the mean values of the balance voltage and the fall time.

For the first oil droplet

$$\overline{V_1} = \frac{V_{11}+V_{12}+V_{13}}{3} = \frac{144V+151V+150V}{3} = 146\text{V}$$

$$\overline{t_1} = \frac{t_{11}+t_{12}+t_{13}}{3} = \frac{38.0V+38.3V+38.3V}{3} = 38.3V$$

Compute the averaged values of the ten oil droplets and put them in Data Table 4-1.

(2) Use the following equation to calculate the charges carried by an oil droplet. Show the sample calculation for the first droplet and list all the charges in Table 4-2.

$$\bar{q} = ne = \frac{1.43 \times 10^{-14}}{[\bar{t}(1+0.02\sqrt{\bar{t}})]^{\frac{3}{2}}} \cdot \frac{1}{\bar{V}}$$

Table. 4-2 the charges and the number of excess electrons on measured droplets

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Oil droplets	Charges $\bar{q}$ (C)	the number of excess electrons, n	Elementary charge e(C)
1	3.422	2	1.71
2	5.898	4	1.47
3	4.213	3	1.40
4	7.875	5	1.58
5	4.448	3	1.48
6	3.405	2	1.70
7	5.056	3	1.69
8	4.995	3	1.67
9	4.717	3	1.57
10	3.506	2	1.75
		Averaged	1.602

(3) Compute the number of excess electrons for every droplet. Suppose that the elementary charge e=1.602×10^{-19} C and the number of excess electrons is

$$n = [\frac{q}{e}]$$

("[ ]" means rounding the quotient to an integer)

Show the sample calculation for the first oil droplet and put all the results in Table 4-2.

(4) Compute the elementary charge.

$$e = \frac{\bar{q}}{n}$$

Show the sample calculation for the first oil droplet and put all the results in Table 4-2.

(5) Compute the averaged elementary charge. Put the final result in Table 4-2.

$$\bar{e} = \sum_{i=1}^{10} e_i$$

(6) Compute the relative error of the elementary charge.

$$E = \frac{\bar{e}-e}{e} \times 100\% = \frac{1.602\times10-19\ C-1.602\times10-19\ C}{1.602\times10-19\ C} = 0$$

(e=1.602×10^{-19} C)

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Conclusions (About 100 words, 5 points)

To sum up, the experiment employing the Millikan oil drop appratus has provided comprehensive insights into the motion of the electron under the influence of electric field force and the viscous resistance. Through numerable measurement , we calculate the charges oil drop brought and the number of electrons and furthermore the value of elementary electron. After calculation, the average value of elementary electron is 1.602×10^{-19} C.

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Answers to Postlab Questions (10 points)

1.Some oil droplets are too big so that their radiums are big,because $r^2 = \frac{9\eta v}{2\varrho g}$ so their final speed is big.\

2.no , they are too small.

3. Coulomb law F=$\frac{kq_1q_2}{r^2}$ if we know the value of an elementary electron ,

Then K can be calculated.

3Appendix

(Scanned data sheets)

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3.11.5 Experimental Data

Data Table 3.11-1 ***Purpose*: To measure the electric charges carried by an oil droplet**

Oil droplets	Balance voltage *V*/V		Fall time *t*/s	
	Measurement	Average	Measurement	Average
1	144 151 150	146	38.0 38.6 38.3	38.3
2	282 281 281	281	17.8 18.2 18.0	18.0
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8	256 257 258	257	21.0 21.2 21.4	21.2
9	179 176 185	180	27.8 27.4 27.6	27.6
10	222 224 227	225	29.2 28.4 29.3	29.0

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